```
!pip install -q vllm
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.0/44.0 kB 2.3 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 87.9/87.9 kB 5.7 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 432.3/432.3 MB 3.3 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 196.5/196.5 kB 10.8 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 45.5/45.5 kB 2.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 267.7/267.7 MB 4.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7.8/7.8 MB 78.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 111.0/111.0 kB 10.6 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 45.4/45.4 kB 5.1 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.9/3.9 MB 84.8 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.3/2.3 MB 85.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 627.7/627.7 kB 43.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 114.2/114.2 kB 12.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.0/3.0 MB 91.5 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.5/6.5 MB 78.1 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 121.4/121.4 kB 13.5 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.2/2.2 MB 50.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 74.4/74.4 MB 9.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.0/3.0 MB 55.3 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 72.1/72.1 kB 3.8 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 180.2/180.2 kB 9.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 69.0/69.0 kB 5.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 231.6/231.6 kB 15.6 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 232.1/232.1 kB 21.0 MB/s eta
0:00:00
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.1/1.1 MB 35.3 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.0/12.0 MB 78.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 42.2/42.2 MB 13.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 388.0/388.0 kB 22.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 288.2/288.2 kB 23.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 149.7/149.7 kB 14.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 225.0/225.0 kB 12.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 180.7/180.7 kB 14.2 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 71.6/71.6 kB 6.2 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.6/2.6 MB 86.7 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 566.4/566.4 kB 26.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 42.4/42.4 MB 13.2 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 79.5/79.5 kB 3.8 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 320.7/320.7 kB 11.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 61.6/61.6 kB 4.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 897.8/897.8 kB 30.5 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 331.1/331.1 kB 15.2 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 8.0/8.0 MB 24.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 819.4/819.4 kB 15.7 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 959.8/959.8 kB 15.1 MB/s eta
0:00:00
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
ipython 7.34.0 requires jedi>=0.16, which is not installed.
google-adk 1.28.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you
have opentelemetry-api 1.41.0 which is incompatible.
google-adk 1.28.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you
have opentelemetry-sdk 1.41.0 which is incompatible.
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-
```

```
sdk<1.39.0,>=1.35.0, but you have opentelemetry-sdk 1.41.0 which is
incompatible.
```

```python
# 2. Import vLLM
from vllm import LLM, SamplingParams
import time

# 3. Define prompts
prompts = [
    "The sun seems warmer today",
    "I hope we can go out ",
    "Are you listening?",
]

# 4. Define sampling parameters
sampling_params = SamplingParams(
    temperature=0.8,
    top_p=0.95,
    max_tokens=50,  # number of tokens to generate
)

# 5. Run generation
llm1 = LLM(model="TinyLlama/TinyLlama-1.1B-Chat-v1.0")

print("Generating completions with TinyLlama (vLLM)...")
start_time_tinyllama = time.time()
outputs_tinyllama = llm1.generate(prompts, sampling_params)
end_time_tinyllama = time.time()
tinyllama_inference_time = end_time_tinyllama - start_time_tinyllama

# 6. Store and Print results
tinyllama_completions = []
for i, output in enumerate(outputs_tinyllama):
    print(f"\nPrompt {i+1}: {prompts[i]}")
    print(f"Completion (TinyLlama): {output.outputs[0].text}")
    tinyllama_completions.append(output.outputs[0].text)

print(f"\nTinyLlama Inference Time: {tinyllama_inference_time:.2f}
seconds")
```

```
INFO 04-14 16:18:04 [utils.py:233] non-default args:
{'disable_log_stats': True, 'model': 'TinyLlama/TinyLlama-1.1B-Chat-
v1.0'}
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:104: UserWarning:
Error while fetching `HF_TOKEN` secret value from your vault:
'Requesting secret HF_TOKEN timed out. Secrets can only be fetched
when running from the Colab UI.'.
You are not authenticated with the Hugging Face Hub in this notebook.
If the error persists, please let us know by opening an issue on
```

```
GitHub (https://github.com/huggingface/huggingface_hub/issues/new).
  warnings.warn(
```

{"model_id":"44496d7b82884ca18c63d5d29f6805f9","version_major":2,"version_minor":0}

```
INFO 04-14 16:18:39 [model.py:549] Resolved architecture:
LlamaForCausalLM
WARNING 04-14 16:18:39 [model.py:1963] Your device 'Tesla T4' (with
compute capability 7.5) doesn't support torch.bfloat16. Falling back
to torch.float16 for compatibility.
WARNING 04-14 16:18:39 [model.py:2016] Casting torch.bfloat16 to
torch.float16.
INFO 04-14 16:18:39 [model.py:1678] Using max model len 2048
INFO 04-14 16:18:39 [scheduler.py:238] Chunked prefill is enabled with
max_num_batched_tokens=8192.
INFO 04-14 16:18:39 [vllm.py:790] Asynchronous scheduling is enabled.
```

{"model_id":"98fd3026f48a4e7fb8d0bf775c8d0763","version_major":2,"version_minor":0}

{"model_id":"6f8ccafa35134cb59902f10e03c8d04e","version_major":2,"version_minor":0}

{"model_id":"73226c510dde457b99c9754fa15ae36a","version_major":2,"version_minor":0}

{"model_id":"c0b6dcc0669243fd92e6d2173f8df1cc","version_major":2,"version_minor":0}

{"model_id":"444af76d337f489ba77cbcd45d9c11d3","version_major":2,"version_minor":0}

```
WARNING 04-14 16:18:41 [system_utils.py:152] We must use the `spawn`
multiprocessing start method. Overriding VLLM_WORKER_MULTIPROC_METHOD
to 'spawn'. See
https://docs.vllm.ai/en/latest/usage/troubleshooting.html#python-
multiprocessing for more information. Reasons: CUDA is initialized
Generating completions with TinyLlama (vLLM)...
```

{"model_id":"108e46c7571d41c58acffd5c50df0839","version_major":2,"version_minor":0}

{"model_id":"4217ced7873d480584bcd2695bfd9a89","version_major":2,"version_minor":0}

```
Prompt 1: The sun seems warmer today
Completion (TinyLlama):
2. And the trees are a lush green
3. The flowers are in full bloom
4. Birds are singing their melodies
```

```
5. The sky is a canvas painted with shades of blue
6. The

Prompt 2: I hope we can go out
Completion (TinyLlama): 1 or 2 more times this year.
Can you tell me more about the activities available during the
Christmas carol concert and how much they cost?

Prompt 3: Are you listening?
Completion (TinyLlama):

JEN: (sarcastically) That's not what I meant. (pauses) Yeah, I heard
that.

KAREN: (smiling) Cool, because now I'm excited to

TinyLlama Inference Time: 1.62 seconds
```

```python
from transformers import AutoTokenizer, AutoModelForCausalLM
import time

# Load EleutherAI/pythia-1b using Hugging Face Transformers
print("\nLoading EleutherAI/pythia-1b using Transformers...")
tokenizer = AutoTokenizer.from_pretrained("EleutherAI/pythia-1b")
model = AutoModelForCausalLM.from_pretrained("EleutherAI/pythia-1b")

# Generate completions for each prompt
print("Generating completions with Pythia-1b...")
pythia_completions = []
start_time_pythia = time.time()
for i, prompt in enumerate(prompts):
    inputs = tokenizer(prompt, return_tensors="pt")

    # Generate output based on sampling_params
    # Adapt vLLM's SamplingParams to Hugging Face generate parameters
    outputs = model.generate(
        **inputs,
        max_new_tokens=sampling_params.max_tokens,
        do_sample=True,
        temperature=sampling_params.temperature,
        top_p=sampling_params.top_p
    )

    # Decode and print the completion
    completion = tokenizer.decode(outputs[0],
skip_special_tokens=True)
    print(f"\nPrompt {i+1}: {prompt}")
    print(f"Completion (Pythia-1b): {completion}")
    pythia_completions.append(completion)
end_time_pythia = time.time()
```

```
pythia_inference_time = end_time_pythia - start_time_pythia

print(f"\nPythia-1b Inference Time: {pythia_inference_time:.2f}
seconds")
```

```
Loading EleutherAI/pythia-1b using Transformers...

{"model_id":"50a730c8afa64923b0864f5ebc883d1d","version_major":2,"vers
ion_minor":0}

{"model_id":"b5655464790f4d398f75b21871dd3bef","version_major":2,"vers
ion_minor":0}

{"model_id":"45f781e265e34b6487337cdf358e63d5","version_major":2,"vers
ion_minor":0}

{"model_id":"62d2b3b98b9c493fbb8de729e4051b35","version_major":2,"vers
ion_minor":0}

{"model_id":"cefad6988dd44036899887b20c74f400","version_major":2,"vers
ion_minor":0}

Setting `pad_token_id` to `eos_token_id`:0 for open-end generation.

Generating completions with Pythia-1b...

Setting `pad_token_id` to `eos_token_id`:0 for open-end generation.


Prompt 1: The sun seems warmer today
Completion (Pythia-1b): The sun seems warmer today, and the air is
cleaner. The air feels
more comfortable and the air feels better than it did before.

Thats all that really matters. It is not the cold, it is the
comfort. The cold is only a small

Setting `pad_token_id` to `eos_token_id`:0 for open-end generation.


Prompt 2: I hope we can go out
Completion (Pythia-1b): I hope we can go out
of this meeting with some positive results.

Vince

Prompt 3: Are you listening?
Completion (Pythia-1b): Are you listening? I can't hear you."

"We're fine, and there are no more questions," said the woman. "Just
let me know when you're ready."
```

```
"Let me know when you're ready?"

"Yes,"

Pythia-1b Inference Time: 46.09 seconds
```

```python
# Load anas-awadalla/mpt-1b-redpajama-200b using Hugging Face
Transformers
from transformers import AutoModelForCausalLM, AutoTokenizer
import time

print("\nLoading anas-awadalla/mpt-1b-redpajama-200b using
Transformers...")
tokenizer_mpt = AutoTokenizer.from_pretrained("anas-awadalla/mpt-1b-
redpajama-200b", trust_remote_code=True)
model_mpt = AutoModelForCausalLM.from_pretrained("anas-awadalla/mpt-
1b-redpajama-200b", trust_remote_code=True, dtype="auto")

# Generate completions for each prompt
print("Generating completions with MPT model...")
mpt_completions = []
start_time_mpt = time.time()
for i, prompt in enumerate(prompts):
    inputs_mpt = tokenizer_mpt(prompt, return_tensors="pt")

    # Generate output based on sampling_params
    outputs_mpt = model_mpt.generate(
        **inputs_mpt,
        max_new_tokens=sampling_params.max_tokens,
        do_sample=True,
        temperature=sampling_params.temperature,
        top_p=sampling_params.top_p
    )

    # Decode and print the completion
    completion_mpt = tokenizer_mpt.decode(outputs_mpt[0],
skip_special_tokens=True)
    print(f"\nPrompt {i+1}: {prompt}")
    print(f"Completion (MPT): {completion_mpt}")
    mpt_completions.append(completion_mpt)
end_time_mpt = time.time()
mpt_inference_time = end_time_mpt - start_time_mpt

print(f"\nMPT Inference Time: {mpt_inference_time:.2f} seconds")
```

```
Loading anas-awadalla/mpt-1b-redpajama-200b using Transformers...
```

{"model_id":"2a788656e1ea4346957c140fcb28cfb9","version_major":2,"version_minor":0}

{"model_id":"b60cd64f5623430291d7794be8b6efad","version_major":2,"version_minor":0}

{"model_id":"6d973a2105e748b8818ad020fdf22f2d","version_major":2,"version_minor":0}

{"model_id":"d04815ad36944997baafd2251214f5c3","version_major":2,"version_minor":0}

{"model_id":"47122809edab4de6838508399e68ddec","version_major":2,"version_minor":0}

```
A new version of the following files was downloaded from
https://huggingface.co/anas-awadalla/mpt-1b-redpajama-200b:
- configuration_mosaic_gpt.py
. Make sure to double-check they do not contain any added malicious
code. To avoid downloading new versions of the code file, you can pin
a revision.
```

{"model_id":"dc3b2eda2f684a75972b3033414c4aaf","version_major":2,"version_minor":0}

{"model_id":"4ca6ad8245c44cd4b3b2ba64cd84359b","version_major":2,"version_minor":0}

```
A new version of the following files was downloaded from
https://huggingface.co/anas-awadalla/mpt-1b-redpajama-200b:
- low_precision_layernorm.py
. Make sure to double-check they do not contain any added malicious
code. To avoid downloading new versions of the code file, you can pin
a revision.
```

{"model_id":"cb31af56e11c4c9d8315051c05e29d08","version_major":2,"version_minor":0}

```
A new version of the following files was downloaded from
https://huggingface.co/anas-awadalla/mpt-1b-redpajama-200b:
- attention.py
. Make sure to double-check they do not contain any added malicious
code. To avoid downloading new versions of the code file, you can pin
a revision.
```

{"model_id":"d769836618e94dd692832900ed0e25a3","version_major":2,"version_minor":0}

```
A new version of the following files was downloaded from
https://huggingface.co/anas-awadalla/mpt-1b-redpajama-200b:
- param_init_fns.py
. Make sure to double-check they do not contain any added malicious
code. To avoid downloading new versions of the code file, you can pin
a revision.
```

{"model_id":"9906b26dc6e94570b4f8557c23f409a2","version_major":2,"version_minor":0}

```
A new version of the following files was downloaded from
https://huggingface.co/anas-awadalla/mpt-1b-redpajama-200b:
- gpt_blocks.py
. Make sure to double-check they do not contain any added malicious
code. To avoid downloading new versions of the code file, you can pin
a revision.
A new version of the following files was downloaded from
https://huggingface.co/anas-awadalla/mpt-1b-redpajama-200b:
- mosaic_gpt.py
- low_precision_layernorm.py
- attention.py
- param_init_fns.py
- gpt_blocks.py
. Make sure to double-check they do not contain any added malicious
code. To avoid downloading new versions of the code file, you can pin
a revision.
```

{"model_id":"c6e65e6abf6242b9ac67fbee7b765c0a","version_major":2,"version_minor":0}

```
/root/.cache/huggingface/modules/transformers_modules/
anas_hyphen_awadalla/mpt_hyphen_1b_hyphen_redpajama_hyphen_200b/
50d6bc94e17812873f39c36c5f815263fa71fb80/attention.py:289:
UserWarning: Using `attn_impl: torch`. If your model does not use
`alibi` or `prefix_lm` we recommend using `attn_impl: flash` otherwise
we recommend using `attn_impl: triton`.
  warnings.warn(
/root/.cache/huggingface/modules/transformers_modules/anas_hyphen_awad
alla/mpt_hyphen_1b_hyphen_redpajama_hyphen_200b/
50d6bc94e17812873f39c36c5f815263fa71fb80/param_init_fns.py:45:
UserWarning: Using a non-tuple sequence for multidimensional indexing
is deprecated and will be changed in pytorch 2.9; use x[tuple(seq)]
instead of x[seq]. In pytorch 2.9 this will be interpreted as tensor
index, x[torch.tensor(seq)], which will result either in an error or a
different result (Triggered internally at
/pytorch/torch/csrc/autograd/python_variable_indexing.cpp:347.)
  init_fn_(module.weight[slice_indices])  # type: ignore

You are using config.init_device='cpu', but you can also use
config.init_device="meta" with Composer + FSDP for fast
initialization.
```

{"model_id":"bb6a0875f2e24be4b6b01ab8f581e066","version_major":2,"version_minor":0}

```
Generating completions with MPT model...

Prompt 1: The sun seems warmer today
```

```
Completion (MPT): The sun seems warmer today, and so does the world.
“The sun seems warmer today, and so does the world.”
— Thomas Moore
The sun is the source of all warmth on Earth.
And it’s not a metaphor.
The sun is

Prompt 2: I hope we can go out
Completion (MPT): I hope we can go out
with a bang, that is the question.
Agree to disagree, agree to disagree.
It's no secret
I think it's good to disagree.
It's good to disagree
when you can't have

Prompt 3: Are you listening?
Completion (MPT): Are you listening?
Pandemic of the mind
The pandemic of the mind has been described as a ‘pandemic of the
body’ — the body in terms of the ‘body politic’. The body politic is
the political domain of the body — the legislature

MPT Inference Time: 2542.37 seconds
```

```python
import pandas as pd

# Create a DataFrame for comparison
comparison_df = pd.DataFrame({
    'Prompt': prompts,
    'TinyLlama (vLLM)': tinyllama_completions,
    'Pythia-1b (Transformers)': pythia_completions,
    'MPT-1b (Transformers)': mpt_completions
})

print("\n--- Model Completion Comparison ---")
display(comparison_df)

print("\n--- Model Inference Time Comparison ---")
print(f"TinyLlama Inference Time: {tinyllama_inference_time:.2f}
seconds")
print(f"Pythia-1b Inference Time: {pythia_inference_time:.2f}
seconds")
print(f"MPT-1b Inference Time: {mpt_inference_time:.2f} seconds")
```

```
--- Model Completion Comparison ---
```


```
{"summary":"{\n  \"name\": \"comparison_df\",\n  \"rows\": 3,\n
\"fields\": [\n  {\n  \"column\": \"Prompt\",\n
\"properties\": {\n  \"dtype\": \"string\",\n
```

\"num_unique_values\": 3,\n   \"samples\": [\n   \"The sun seems warmer today\",\n   \"I hope we can go out \",\n   \"Are you listening?\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"TinyLlama (vLLM)\",\n   \"properties\": {\n   \"dtype\": \"string\",\n   \"num_unique_values\": 3,\n   \"samples\": [\n   \"\\n2. And the trees are a lush green\\n3. The flowers are in full bloom\\n4. Birds are singing their melodies\\n5. The sky is a canvas painted with shades of blue\\n6. The\",\n   \"1 or 2 more times this year.\\nCan you tell me more about the activities available during the Christmas carol concert and how much they cost?\",\n   \"\\n\\nJEN: (sarcastically) That's not what I meant. (pauses) Yeah, I heard that.\\n\\nKAREN: (smiling) Cool, because now I'm excited to\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"Pythia-1b (Transformers)\",\n   \"properties\": {\n   \"dtype\": \"string\",\n   \"num_unique_values\": 3,\n   \"samples\": [\n   \"The sun seems warmer today, and the air is cleaner. The air feels\\nmore comfortable and the air feels better than it did before.\\n\\nThats all that really matters. It is not the cold, it is the\\ncomfort. The cold is only a small\",\n   \"I hope we can go out \\nof this meeting with some positive results.\\n\\nVince\",\n   \"Are you listening? I can't hear you.\\\"\\n\\n\\\"We're fine, and there are no more questions,\\\" said the woman. \\\"Just let me know when you're ready.\\\"\\n\\n\\\"Let me know when you're ready?\\\"\\n\\n\\\"Yes,\\\"\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"MPT-1b (Transformers)\",\n   \"properties\": {\n   \"dtype\": \"string\",\n   \"num_unique_values\": 3,\n   \"samples\": [\n   \"The sun seems warmer today, and so does the world.\\n\\u201cThe sun seems warmer today, and so does the world.\\u201d\\n\\u2014 Thomas Moore\\nThe sun is the source of all warmth on Earth.\\nAnd it\\u2019s not a metaphor.\\nThe sun is\",\n   \"I hope we can go out \\nwith a bang, that is the question. \\nAgree to disagree, agree to disagree. \\nIt's no secret \\nI think it's good to disagree.\\nIt's good to disagree \\nwhen you can't have\",\n   \"Are you listening?\\nPandemic of the mind\\nThe pandemic of the mind has been described as a \\u2018pandemic of the body\\u2019 \\u2013 the body in terms of the \\u2018body politic\\u2019. The body politic is the political domain of the body \\u2013 the legislature\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe","variable_name":"comparison_df"}

```
--- Model Inference Time Comparison ---
TinyLlama Inference Time: 1.62 seconds
Pythia-1b Inference Time: 46.09 seconds
MPT-1b Inference Time: 2542.37 seconds
```